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MIL-M-28787/463
22 May 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SEMICONDUCTOR MEMORY ARRAY, STATIC RAM, 256K MODULE,

KEY CODE JMJ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic 256K, 22-bit word semiconductor static random access memory (SRAM) module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 98.548 failures per million operating hours for class I or 417.421 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be JMJ.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2-A21	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B7	Continuity	III	X	0	1	0	1	0	1	Ohm
C1,C2	Vpp stress	IV	0	---------- Module shall pass all remaining tests ----------						
D1-D10	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D11,D12	Ibd	V	S	-0.005	0.005	-0.005	0.005	-0.005	0.005	mA
D13	Ibd	V	S	-0.61	0.03	-0.61	0.03	-0.61	0.03	mA
D14	Ibd	V	S	-0.005	0.005	-0.005	0.005	-0.005	0.005	mA
D15-D22	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D23	Ibd	V	S	-0.03	0.03	-0.03	0.03	-0.03	0.03	mA
D24	Ibd	V	S	-0.058	0.158	-0.058	0.158	-0.058	0.158	mA
D25-D42	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
E1-E22	Ios	VI	S	-350.0	-80.0	-350.0	-80.0	-350.0	-80.0	nA
F1-F4	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F5-F8	Iih	VII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
F9-F14	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F15-F19	Iih	VII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
F19,F20	Iih	VII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	mA
F21	Iih	VII	X	-0.61	0.03	-0.61	0.03	-0.61	0.03	mA
F22	Iih	VII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	mA
F23-F25	Iih	VII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
F26-F29	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F30-F33	Iih	VII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
F34-F37	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F38-F41	Iih	VII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
F42	Iih	VII	X	-0.03	0.03	-0.03	0.03	-0.03	0.03	mA
F43	Iih	VII	X	-3.32	-1.42	-3.32	-1.42	-3.32	-1.42	mA
F44-F46	Iih	VII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
F47-F64	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 50 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G1-G4	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G5-G8	Iil	VIII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
G9-G14	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G15-G18	Iil	VIII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
G19,G20	Iil	VIII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	mA
G21	Iil	VIII	X	-0.61	0.03	-0.61	0.03	-0.61	0.03	mA
G22	Iil	VIII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	mA
G23-G25	Iil	VIII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
G26-G29	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G30-G33	Iil	VIII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
G34-G37	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G38-G41	Iil	VIII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
G42	Iil	VIII	X	-0.03	0.03	-0.03	0.03	-0.03	0.03	mA
G43	Iil	VIII	X	-3.32	-1.42	-3.32	-1.42	-3.32	-1.42	mA
G44-G46	Iil	VIII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	mA
G47-G84	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
H1-H22	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J25	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K4	Iozh	XI	X	-0.01	0.01	-0.01	0.01	-0.01	0.01	mA
K5-K8	Iozl	XI	X	-0.01	0.01	-0.01	0.01	-0.01	0.01	mA
L1-L10	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L11-L14	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L15-L32	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L33	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L34-L41	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
M1,M2	PBFC	XIII	S	2.8	5.2	2.8	5.2	2.8	5.2	uF
M3	PBFC	XIII	S	7.4	13.8	7.4	13.8	7.4	13.8	uF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 80 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
N1	Ieo	XIV	X	100	300	100	300	100	300	mA
N2	Iec	XIV	X	200	1500	200	1500	200	1500	mA
N3	Ice	XIV	X	0	20	0	20	0	20	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1-R12	Cin	XVIII	Q	2	50	2	50	2	50	pF
R13	Cin	XVIII	Q	4	75	4	75	4	75	pF
R14-R32	Cin	XVIII	Q	2	50	2	50	2	50	pF
R33	Cin	XVIII	Q	6	150	6	150	6	150	pF
R34	Cin	XVIII	Q	4	75	4	75	4	75	pF
R35-R42	Cin	XVIII	Q	2	50	2	50	2	50	pF
R43-R64	Cin	XVIII	Q	4	50	4	50	4	50	pF
T1-T3	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/463-1 for class I and M28787/463-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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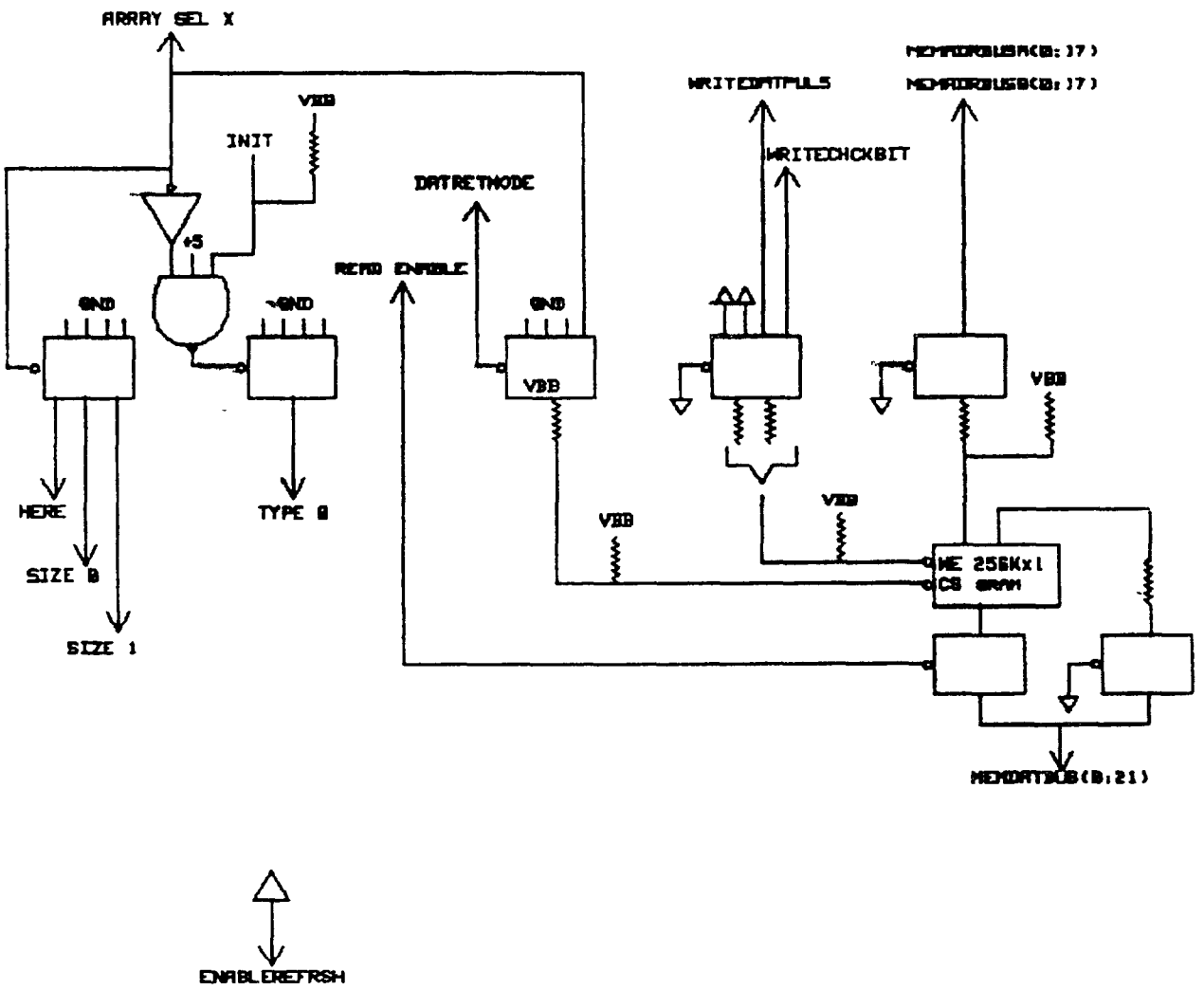


FIGURE 1. JMJ functional block diagram.

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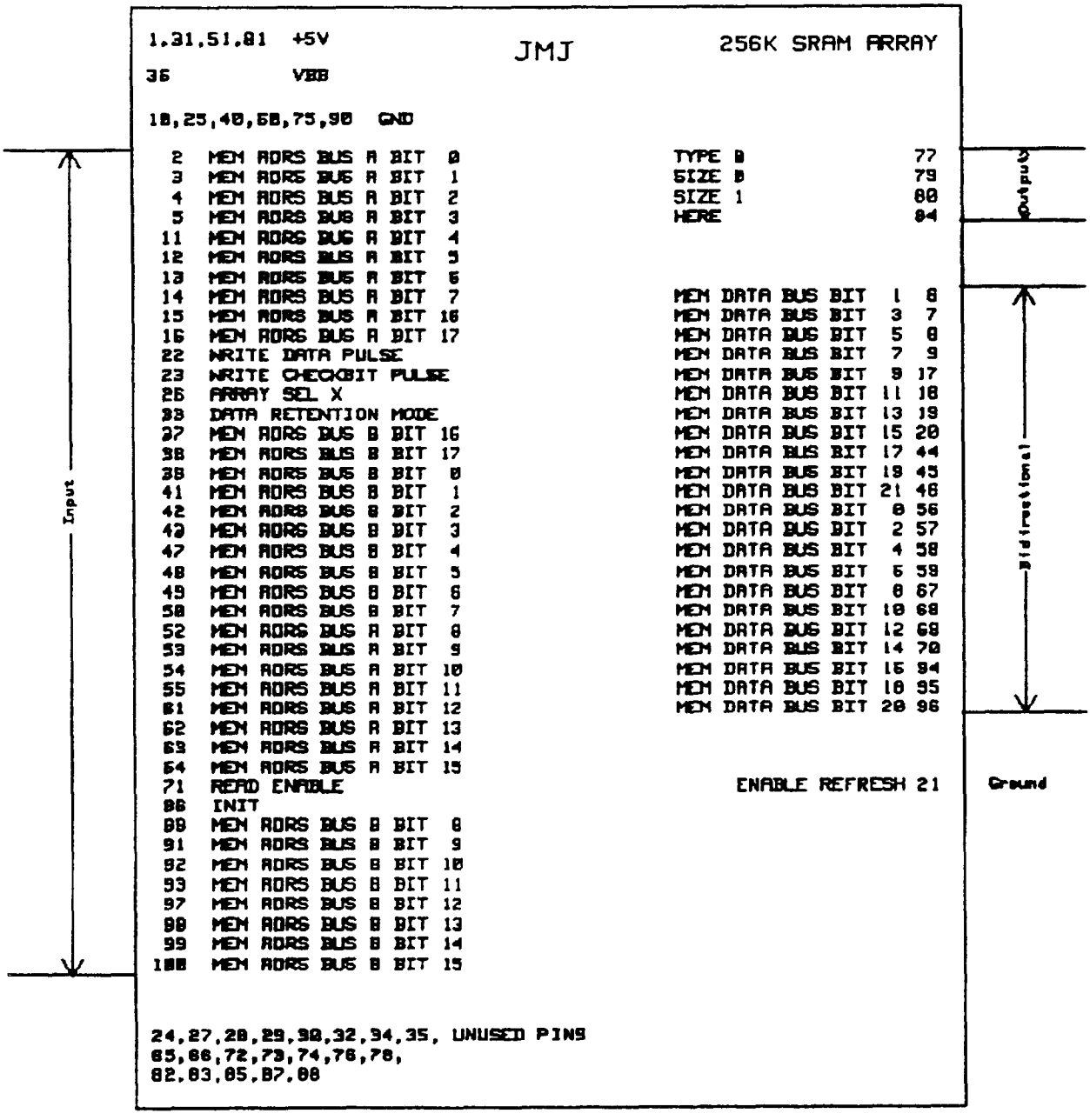


FIGURE 2. JMJ pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

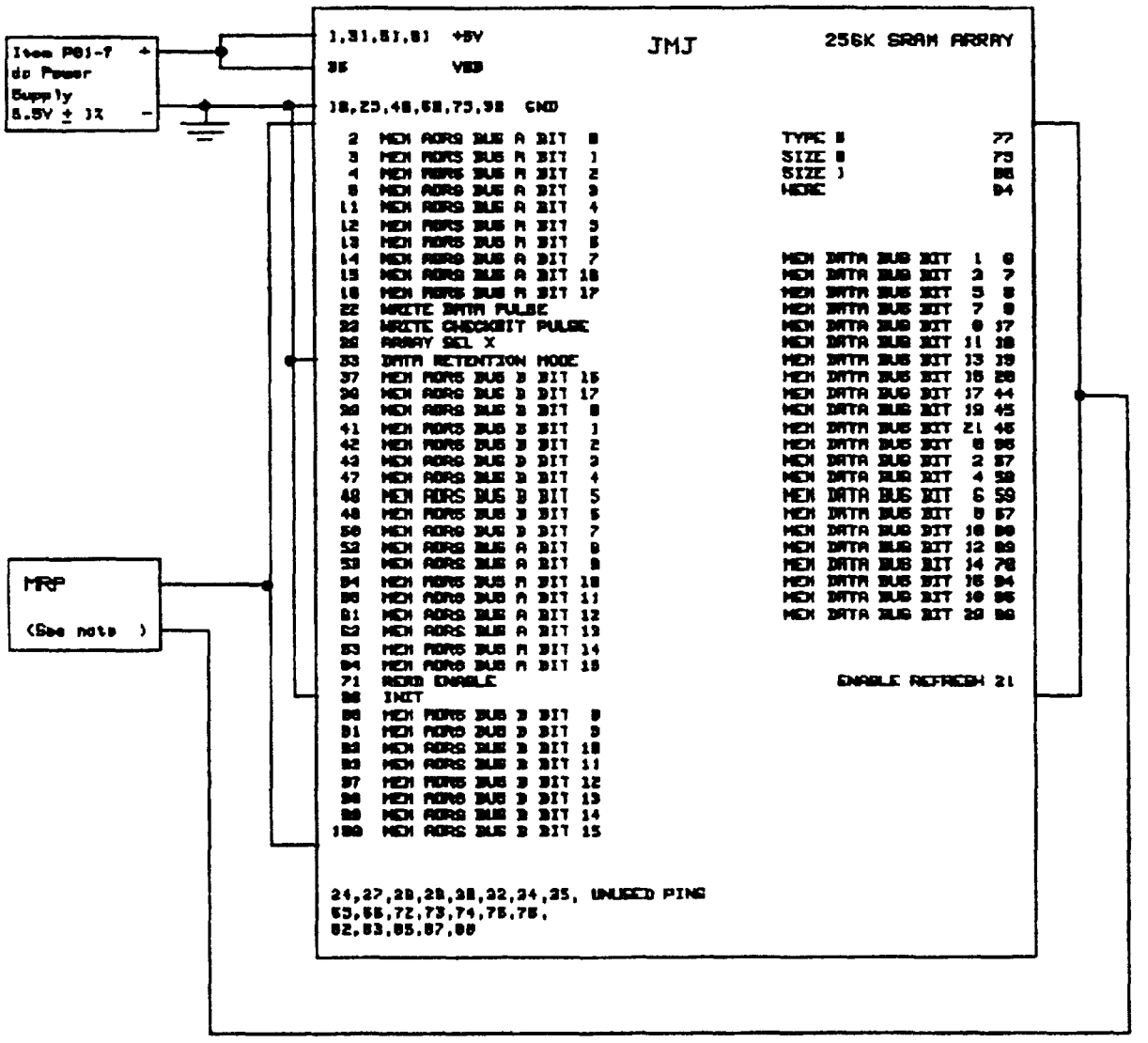
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. JMJ operating temperature and component temperature test circuit.

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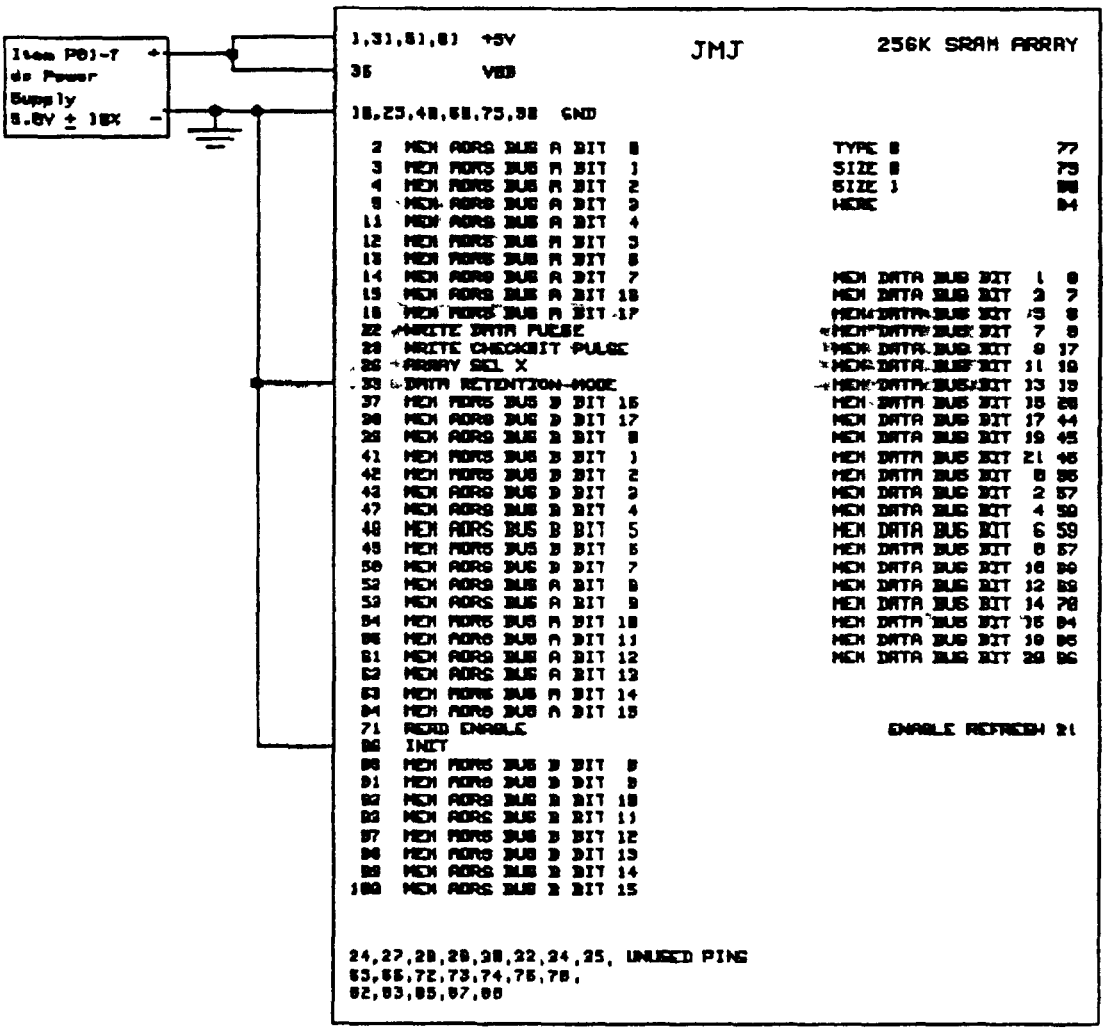


FIGURE 4. JMJ vibration test circuit.

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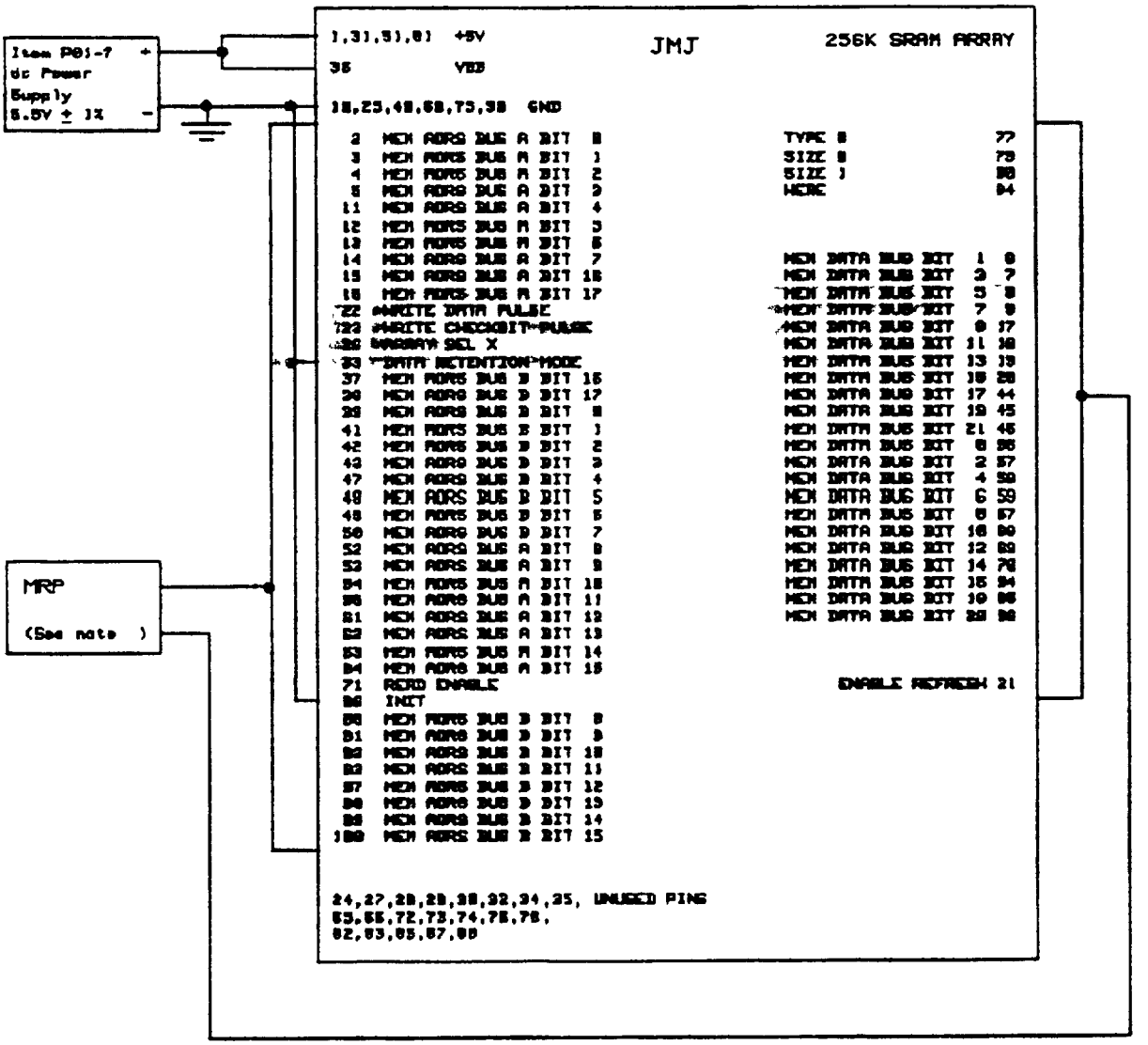
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. JMJ life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 10.01 W (max); 5.68 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEMADRBUSA0	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA1	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA2	-1.5	0.4	2.0	0.02	0.02	0.1	50
MEMADRBUSA3	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA4	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA5	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA6	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA7	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA16	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA17	-1.5	0.4	2.4	0.02	0.02	0.1	50
WRITEDATPULS	-1.2	0.4	2.4	0.005	0.005	0.005	50
WRITECHCKBIT	-1.2	0.4	2.4	0.005	0.005	0.005	50
ARRAY SEL X	-1.2	0.4	2.4	0.03	0.03	0.03	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DATRETMODE	-1.2	0.4	2.4	0.005	0.005	0.005	50
MEMADRBUSB16	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB17	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB0	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB1	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB2	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB3	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB4	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB5	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB6	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB7	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA8	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA9	-1.5	0.4	2.4	0.02	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEMADRBUSA10	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA11	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA12	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA13	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA14	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSA15	-1.5	0.4	2.4	0.02	0.02	0.1	50
READ ENABLE	-1.2	0.4	2.4	0.03	0.03	0.03	150
INIT	NA	0.4	2.4	-1.42	-1.42	0.158	75
MEMADRBUSB8	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB9	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB10	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB11	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB12	-1.5	0.4	2.4	0.02	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEMADRBUSB13	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB14	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMADRBUSB15	-1.5	0.4	2.4	0.02	0.02	0.1	50
MEMDATBUS1	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS3	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS5	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS7	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS9	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS11	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS13	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS15	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS17	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS19	NA	0.4	2.4	0.015	0.015	NA	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MEMDATBUS21	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS0	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS2	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS4	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS6	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS8	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS10	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS12	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS14	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS16	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS18	NA	0.4	2.4	0.015	0.015	NA	50
MEMDATBUS20	NA	0.4	2.4	0.015	0.015	NA	50

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
TYPE 0	NA	0.5	NA	6.0	10	10	NA	NA
SIZE 0	NA	0.5	NA	6.0	10	10	NA	NA
SIZE 1	NA	0.5	NA	6.0	10	10	NA	NA
HERE	NA	0.5	NA	6.0	10	10	NA	NA
MEMDATBUS1	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS3	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS5	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS7	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS9	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS11	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS13	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS15	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS17	2.4	0.5	-0.4	6.0	NA	NA	-60	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
MEMDATBUS19	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS21	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS0	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS2	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS4	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS6	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS8	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS10	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS12	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS14	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS16	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS18	2.4	0.5	-0.4	6.0	NA	NA	-60	NA
MEMDATBUS20	2.4	0.5	-0.4	6.0	NA	NA	-60	NA

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	MEMADRBUSA0 to ARRAY SEL X	10	10	10	-	-	-	nS	12a, 12b
1	tsu	MEMADRBUSA1 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA2 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA3 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA4 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA5 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA6 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA7 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA8 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA9 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA10 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA11 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA12 to ARRAY SEL X	10	10	10	-	-	-	nS	12a, 12b

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	MEMADRBUSA13 to ARRAY SEL X	10	10	10	-	-	-	nS	12a, 12b
1	tsu	MEMADRBUSA14 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA15 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA16 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA17 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB0 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB1 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB2 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB3 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB4 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB5 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB6 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB7 to ARRAY SEL X	10	10	10	-	-	-	nS	12a, 12b

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	MEMADRBUSB8 to ARRAY SEL X	10	10	10	-	-	-	nS	12a, 12b
1	tsu	MEMADRBUSB9 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB10 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB11 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB12 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB13 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB14 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB15 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB16 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB17 to ARRAY SEL X	10	10	10	-	-	-	nS	
2	tsu	ARRAY SEL X to READ ENABLE	85	85	85	-	-	-	nS	
3	tpd	ARRAY SEL X to MEMDATBUS0	-	-	-	180	180	180	nS	12a, 12b

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	ARRAY SEL X to MEMDATBUS1	-	-	-	180	180	180	nS	12a, 12b
3	tpd	ARRAY SEL X to MEMDATBUS2	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS3	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS4	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS5	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS6	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS7	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS8	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS9	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS10	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS11	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS12	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS13	-	-	-	180	180	180	nS	12a, 12b

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	ARRAY SEL X to MEMDATBUS14	-	-	-	180	180	180	nS	12a, 12b
3	tpd	ARRAY SEL X to MEMDATBUS15	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS16	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS17	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS18	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS19	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS20	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS21	-	-	-	180	180	180	nS	
4	tsu	READ ENABLE to MEMADRBUSA0	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA1	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA2	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA3	0	0	0	-	-	-	nS	12a, 12b

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tsu	READ ENABLE to MEMADRBUSA4	0	0	0	-	-	-	nS	12a, 12b
4	tsu	READ ENABLE to MEMADRBUSA5	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA6	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA7	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA8	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA9	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA10	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA11	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA12	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA13	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA14	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA15	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA16	0	0	0	-	-	-	nS	12a, 12b

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tsu	READ ENABLE to MEMADRBUSA17	0	0	0	-	-	-	nS	12a, 12b
4	tsu	READ ENABLE to MEMADRBUSB0	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB1	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB2	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB3	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB4	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB5	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB6	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB7	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB8	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB9	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB10	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB11	0	0	0	-	-	-	nS	12a, 12b

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tsu	READ ENABLE to MEMADRBUSB12	0	0	0	-	-	-	nS	12a, 12b
4	tsu	READ ENABLE to MEMADRBUSB13	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB14	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB15	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB16	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB17	0	0	0	-	-	-	nS	
5	tpw	ARRAY SEL X	156	156	156	-	-	-	nS	
6	tsu	ARRAY SEL X to WRITEDATPULS	83	83	83	-	-	-	nS	
7	tsu	MEMDATBUS0 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS1 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS2 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS3 to WRITEDATPULS	74	74	74	-	-	-	nS	12a, 12b

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tsu	MEMDATBUS4 to WRITEDATPULS	74	74	74	-	-	-	nS	12a, 12b
7	tsu	MEMDATBUS5 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS6 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS7 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS8 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS9 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS10 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tus	MEMDATBUS11 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS12 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS13 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS14 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS15 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS16 to WRITEDATPULS	74	74	74	-	-	-	nS	12a, 12b

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tsu	MEMDATBUS17 to WRITEDATPULS	74	74	74	-	-	-	nS	12a, 12b
7	tsu	MEMDATBUS18 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS19 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS20 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS21 to WRITEDATPULS	74	74	74	-	-	-	nS	
8	tpw	WRITEDATPULS	50	50	50	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA0	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA1	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA2	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA3	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA4	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA5	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA6	8	8	8	-	-	-	ns	12a, 12b

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tsu	WRITEDATPULS to MEMADRBUSA7	8	8	8	-	-	-	nS	12a, 12b
9	tsu	WRITEDATPULS to MEMADRBUSA8	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA9	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA10	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA11	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA12	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA13	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA14	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA15	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA16	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA17	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB0	8	8	8	-	-	-	nS	12a, 12b

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:19Z
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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tsu	WRITEDATPULS to MEMADRBUSB1	8	8	8	-	-	-	nS	12a, 12b
9	tsu	WRITEDATPULS to MEMADRBUSB2	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS tO MEMADRBUSB3	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB4	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB5	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB6	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB7	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB8	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB9	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB10	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB11	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB12	8	8	8	-	-	-	nS	12a, 12b

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tsu	WRITEDATPULS to MEMADRBUSB13	8	8	8	-	-	-	nS	12a, 12b
9	tsu	WRITEDATPULS to MEMADRBUSB14	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB15	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB16	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB17	8	8	8	-	-	-	nS	12a, 12b

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6.4 Module function. The 256K Static Random Access Memory (SRAM) array is divided into two subarrays, one for the even data bits and one for the odd data bits. Each subarray is made up of eleven 256K X 1-bit static Random Access Memory (RAM) integrated circuit devices. The A address bits address odd data bits 1,3,5,7,9,11,13,15,17,19 and 21. The B address bits address even data bits 0,2,4,6,8,10,12,14,16,18 and 20. Data inputs to and outputs from the array are from the 3-state even/odd data buffers. Address inputs are received from the address A and B buffers.

6.4.1 Control feedback circuits. The control feedback circuits are 4-bit buffer circuits with ground strapped inputs. These circuits will provide Here, Size and array Type information when strobed by an active Array Sel X and Init signal gated to the buffer circuits output enable input.

6.4.2 Address A and B buffers. The address A and B buffers provide SRAM isolation from the memory address bus. Outputs are constantly enabled. The A address bits are applied to the odd data bit SRAMs while the B address bits are applied to the even data bit SRAMs.

6.4.3 Write logic. The write logic is a 4-bit buffer circuit which passes the Array Select X, and commands to write data and write checkbits into the SRAMs. The array select signal provides the chip enable for the SRAMs; the write data pulse provides the SRAM write enable; and, the write checkbit pulse is applied to the upper even and odd data SRAM write enable input.

6.4.4 Even/odd data buffers. The even/odd data buffers are 3-state data buffer devices used to provide isolation between the memory data bus and the SRAMs. Data outputs are enabled during a read of the array.

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6.4.5 Description of I/O pins.

Signal name	Pin number	I/O	Description
MEMDATBUS1	6	I/O	MEM DATA BUS BIT 1
MEMDATBUS3	7	I/O	MEM DATA BUS BIT 3
MEMDATBUS5	8	I/O	MEM DATA BUS BIT 5
MEMDATBUS7	9	I/O	MEM DATA BUS BIT 7
MEMDATBUS9	17	I/O	MEM DATA BUS BIT 9
MEMDATBUS11	18	I/O	MEM DATA BUS BIT 11
MEMDATBUS13	19	I/O	MEM DATA BUS BIT 13
MEMDATBUS15	20	I/O	MEM DATA BUS BIT 15
MEMDATBUS17	44	I/O	MEM DATA BUS BIT 17
MEMDATBUS19	45	I/O	MEM DATA BUS BIT 19
MEMDATBUS21	46	I/O	MEM DATA BUS BIT 21
MEMDATBUS0	56	I/O	MEM DATA BUS BIT 0
MEMDATBUS2	57	I/O	MEM DATA BUS BIT 2
MEMDATBUS4	58	I/O	MEM DATA BUS BIT 4
MEMDATBUS6	59	I/O	MEM DATA BUS BIT 6
MEMDATBUS8	67	I/O	MEM DATA BUS BIT 8
MEMDATBUS10	68	I/O	MEM DATA BUS BIT 10
MEMDATBUS12	69	I/O	MEM DATA BUS BIT 12
MEMDATBUS14	70	I/O	MEM DATA BUS BIT 14
MEMDATBUS16	94	I/O	MEM DATA BUS BIT 16

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MEMDATBUS18	95	I/O	MEM DATA BUS BIT 18
MEMDATBUS20	96	I/O	MEM DATA BUS BIT 20
MEMADRBUSA0	2	I	MEM ADRS BUS A BIT 0
MEMADRBUSA1	3	I	MEM ADRS BUS A BIT 1
MEMADRBUSA2	4	I	MEM ADRS BUS A BIT 2
MEMADRBUSA3	5	I	MEM ADRS BUS A BIT 3
MEMADRBUSA4	11	I	MEM ADRS BUS A BIT 4
MEMADRBUSA5	12	I	MEM ADRS BUS A BIT 5
MEMADRBUSA6	13	I	MEM ADRS BUS A BIT 6
MEMADRBUSA7	14	I	MEM ADRS BUS A BIT 7
MEMADRBUSA16	15	I	MEM ADRS BUS A BIT 16
MEMADRBUSA17	16	I	MEM ADRS BUS A BIT 17
WRITEDATPULS	22	I	WRITE DATA PULSE
WRITECHCKBIT	23	I	WRITE CHECKBIT PULSE
ARRAY SEL X	26	I	ARRAY SEL X
DATRETMODE	33	I	DATA RETENTION MODE
MEMADRBUSB16	37	I	MEM ADRS BUS B BIT 16
MEMADRBUSB17	38	I	MEM ADRS BUS B BIT 17
MEMADRBUSB0	39	I	MEM ADRS BUS B BIT 0
MEMADRBUSB1	41	I	MEM ADRS BUS B BIT 1

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MEMADRBUSB2	42	I	MEM ADRS BUS B BIT 2
MEMADRBUSB3	43	I	MEM ADRS BUS B BIT 3
MEMADRBUSB4	47	I	MEM ADRS BUS B BIT 4
MEMADRBUSB5	48	I	MEM ADRS BUS B BIT 5
MEMADRBUSB6	49	I	MEM ADRS BUS B BIT 6
MEMADRBUSB7	50	I	MEM ADRS BUS B BIT 7
MEMADRBUSA8	52	I	MEM ADRS BUS A BIT 8
MEMADRBUSA9	53	I	MEM ADRS BUS A BIT 9
MEMADRBUSA10	54	I	MEM ADRS BUS A BIT 10
MEMADRBUSA11	55	I	MEM ADRS BUS A BIT 11
MEMADRBUSA12	61	I	MEM ADRS BUS A BIT 12
MEMADRBUSA13	62	I	MEM ADRS BUS A BIT 13
MEMADRBUSA14	63	I	MEM ADRS BUS A BIT 14
MEMADRBUSA15	64	I	MEM ADRS BUS A BIT 15
READ ENABLE	71	I	READ ENABLE
INIT	86	I	INIT
MEMADRBUSB8	89	I	MEM ADRS BUS B BIT 8
MEMADRBUSB9	91	I	MEM ADRS BUS B BIT 9
MEMADRBUSB10	92	I	MEM ADRS BUS B BIT 10
MEMADRBUSB11	93	I	MEM ADRS BUS B BIT 11

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MEMADRBUSB12	97	I	MEM ADRS BUS B BIT 12
MEMADRBUSB13	98	I	MEM ADRS BUS B BIT 13
MEMADRBUSB14	99	I	MEM ADRS BUS B BIT 14
MEMADRBUSB15	100	I	MEM ADRS BUS B BIT 15
TYPE 0	77	O	TYPE 0
SIZE 0	79	O	SIZE 0
SIZE 1	80	O	SIZE 1
HERE	84	O	HERE

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0081)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C	Frame	10	INF	MOhm	1-9
A2	Isolation, used to unused pins	C	24,27, 28,29, 30,32, 34,35, 65,66, 72,73, 74,76, 78,82, 83,85, 87,88	10	INF	MOhm	1-9
A3-A21	Isolation, unused to unused pins	see note 4		10	INF	MOhm	1-9

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A21
Isolation measurement	NA	± 500 KOhm	A1-A21

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. The unused pins on this module are: 24, 27, 28, 29, 30, 32, 34, 35, 65, 66, 72, 73, 74, 76, 78, 82, 83, 85, 87, 88.

5. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

6. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

7. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

8. The test circuit shall be as indicated on figure 6.

9. Abbreviations used are defined in 6.1.

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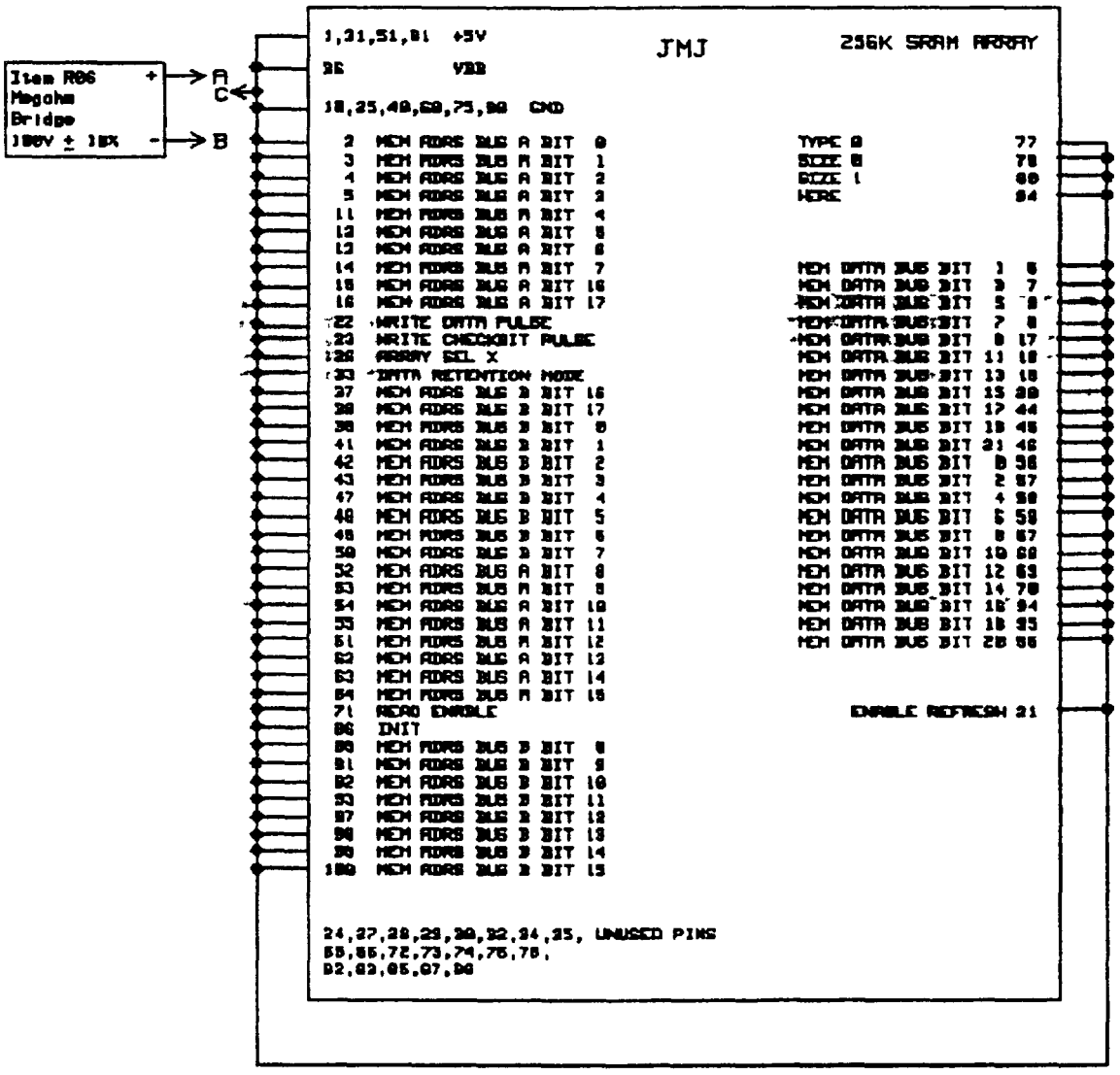


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, GND	10	21				
B6	Continuity, Vcc	1	31				
B7	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B7

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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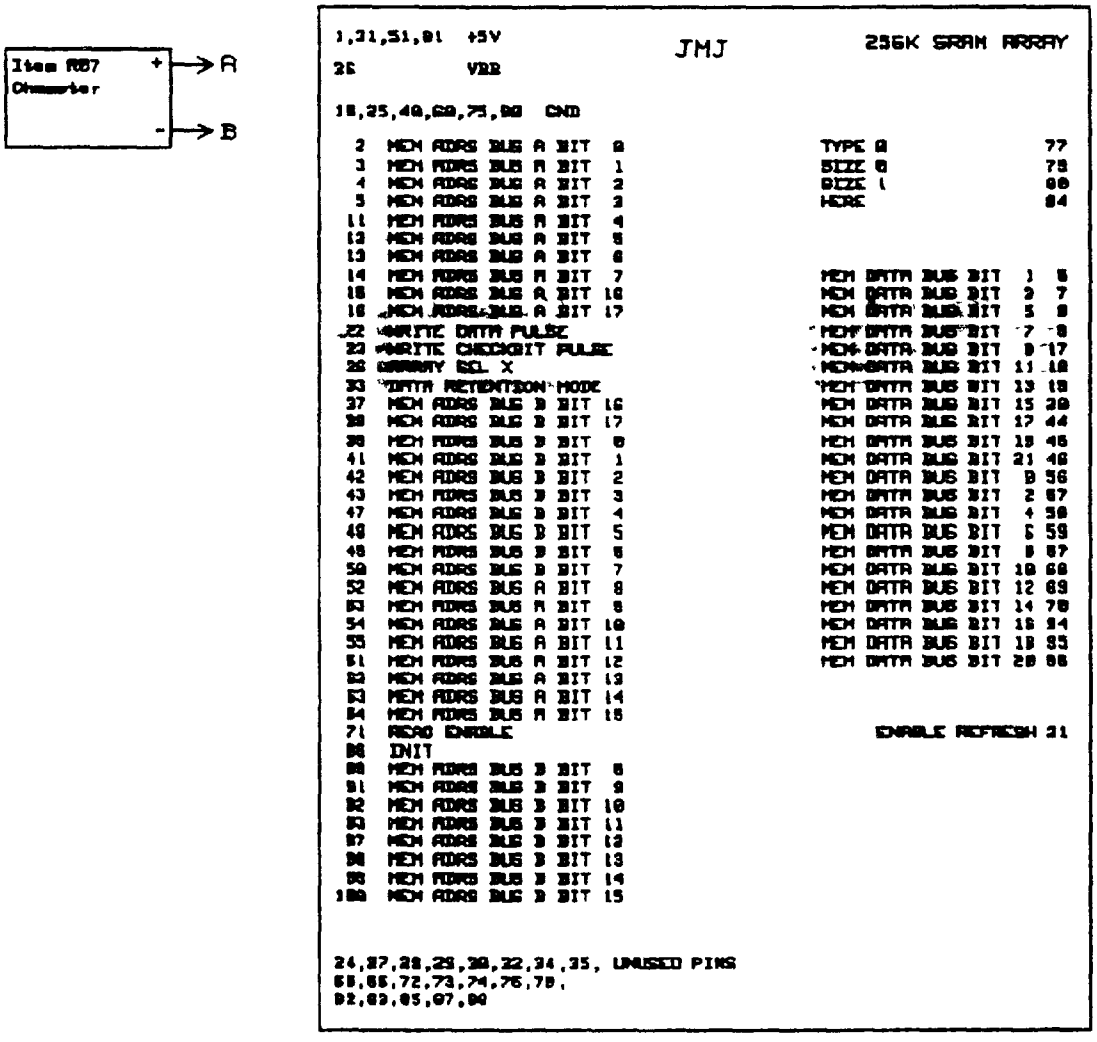


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C1	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90, 33,86	Module shall pass all remaining tests.	1-6
C2	Vbb	7.0	36	10,25, 40,60, 75,90, 33,86	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C1,C2
Power supply current limit	540 mA 2.7 A	NA NA	C1 C2

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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Check the source to verify that this is the current version before use.

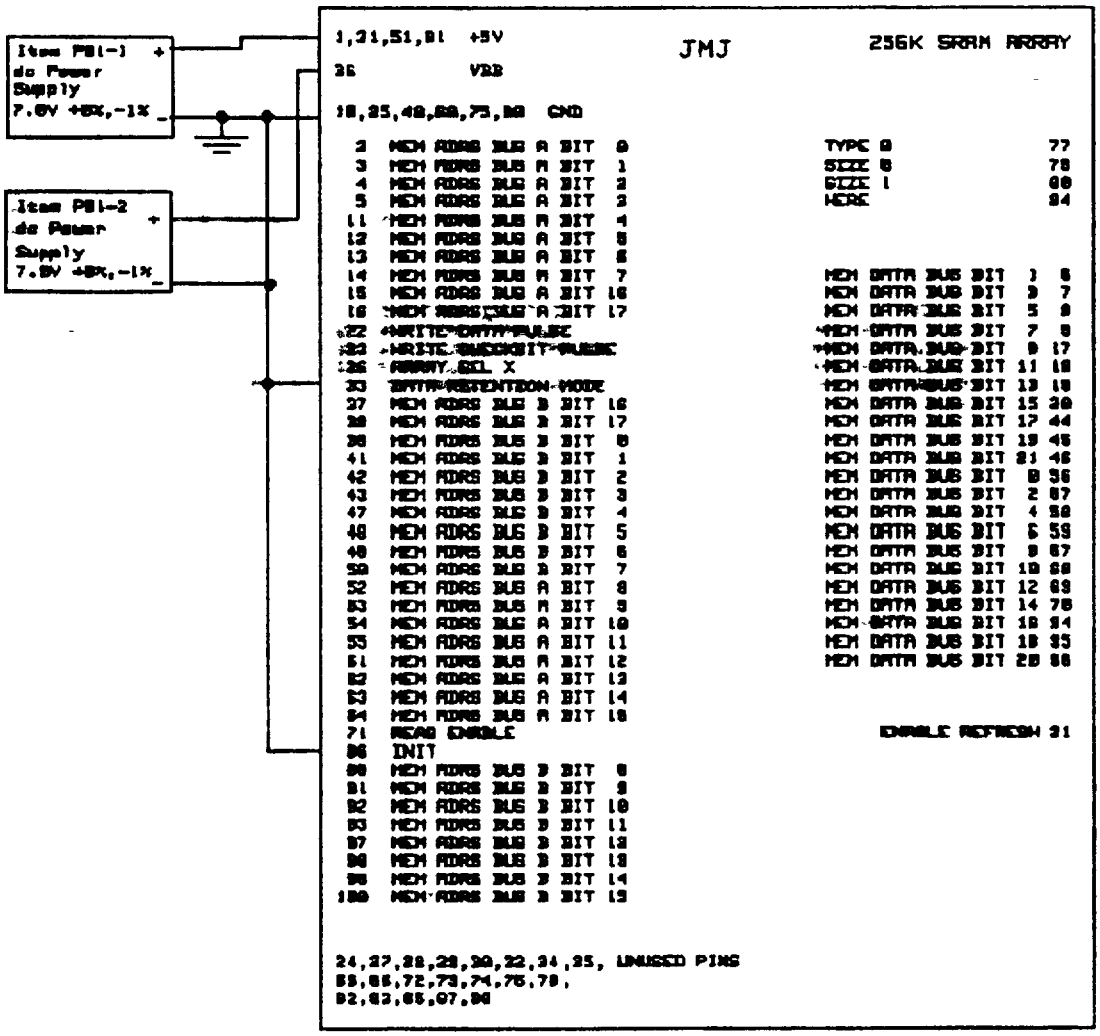


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	MEMADRBUSA0	2	5.5	0.0	0.1	mA	1-3
D2	MEMADRBUSA1	3					
D3	MEMADRBUSA2	4					
D4	MEMADRBUSA3	5					
D5	MEMADRBUSA4	11					
D6	MEMADRBUSA5	12					
D7	MEMADRBUSA6	13					
D8	MEMADRBUSA7	14					
D9	MEMADRBUSA16	15					
D10	MEMADRBUSA17	16	5.5	0.0	0.1	mA	1-3
D11	WRITEDATPULS	22	5.5	-0.005	0.005	mA	1-3
D12	WRITECHCKBIT	23	5.5	-0.005	0.005	mA	1-3
D13	ARRAY SEL X	26	5.5	-0.61	0.03	mA	1-3
D14	DATRETMODE	33	5.5	-0.005	0.005	mA	1-3
D15	MEMADRBUSA8	52	5.5	0.0	0.1	mA	1-3
D16	MEMADRBUSA9	53					
D17	MEMADRBUSA10	54					
D18	MEMADRBUSA11	55					
D19	MEMADRBUSA12	61					
D20	MEMADRBUSA13	62	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	MEMADRBUSA14	63	5.5	0.0	0.1	mA	1-3
D22	MEMADRBUSA15	64	5.5	0.0	0.1	mA	1-3
D23	READ ENABLE	71	5.5	-0.03	0.03	mA	1-3
D24	INIT	86	5.5	-0.058	0.158	mA	1-3
D25	MEMADRBUSB16	37	5.5	0.0	0.1	mA	1-3
D26	MEMADRBUSB17	38					
D27	MEMADRBUSB0	39					
D28	MEMADRBUSB1	41					
D29	MEMADRBUSB2	42					
D30	MEMADRBUSB3	43					
D31	MEMADRBUSB4	47					
D32	MEMADRBUSB5	48					
D33	MEMADRBUSB6	49					
D34	MEMADRBUSB7	50					
D35	MEMADRBUSB8	89					
D36	MEMADRBUSB9	91					
D37	MEMADRBUSB10	92					
D38	MEMADRBUSB11	93					
D39	MEMADRBUSB12	97					
D40	MEMADRBUSB13	98	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	MEMADRBUSB14	99	5.5	0.0	0.1	mA	1-3
D42	MEMADRBUSB15	100	5.5	0.0	0.1	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D42
Power supply current limit	0.45 A	NA	D1-D42
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D42
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D42 D1-D42
Input current measurement	NA NA NA NA NA NA NA NA	± 5.0 uA ± 0.5 uA ± 32.0 uA ± 0.5 uA ± 5.0 uA ± 3.0 uA ± 10.8 uA ± 5.0 uA	D1-D10 D11,D12 D13 D14 D15-D22 D23 D24 D25-D42

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	MEMDATBUS1	6	0.0	-350.0	-60.0	mA	1-4
E2	MEMDATBUS3	7					
E3	MEMDATBUS5	8					
E4	MEMDATBUS7	9					
E5	MEMDATBUS9	17					
E6	MEMDATBUS11	18					
E7	MEMDATBUS13	19					
E8	MEMDATBUS15	20					
E9	MEMDATBUS17	44					
E10	MEMDATBUS19	45					
E11	MEMDATBUS21	46					
E12	MEMDATBUS0	56					
E13	MEMDATBUS2	57					
E14	MEMDATBUS4	58					
E15	MEMDATBUS6	59					
E16	MEMDATBUS8	67					
E17	MEMDATBUS10	68					
E18	MEMDATBUS12	69					
E19	MEMDATBUS14	70					
E20	MEMDATBUS16	94	0.0	-350.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	MEMDATBUS18	95	0.0	-350.0	-60.0	mA	1-4
E22	MEMDATBUS20	96	0.0	-350.0	-60.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E22
Power supply current limit	0.45 A	NA	E1-E22
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E22
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E22 E1-E22
Output current measurement	NA	± 14.5 mA	E1-E22

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	MEMADRBUSA0	2	2.4	-0.2	0.02	mA	1-3
F2	MEMADRBUSA1	3					
F3	MEMADRBUSA2	4					
F4	MEMADRBUSA3	5	2.4	-0.2	0.02	mA	1-3
F5	MEMDATBUS1	6	2.4	-0.015	0.015	mA	1-3
F6	MEMDATBUS3	7					
F7	MEMDATBUS5	8					
F8	MEMDATBUS7	9	2.4	-0.015	0.015	mA	1-3
F9	MEMADRBUSA4	11	2.4	-0.2	0.02	mA	1-3
F10	MEMADRBUSA5	12					
F11	MEMADRBUSA6	13					
F12	MEMADRBUSA7	14					
F13	MEMADRBUSA16	15					
F14	MEMADRBUSA17	16	2.4	-0.2	0.02	mA	1-3
F15	MEMDATBUS9	17	2.4	-0.015	0.015	mA	1-3
F16	MEMDATBUS11	18					
F17	MEMDATBUS13	19					
F18	MEMDATBUS15	20	2.4	-0.015	0.015	mA	1-3
F19	WRITEDATPULS	22	2.4	-0.005	0.005	mA	1-3
F20	WRITECHCKBIT	23	2.4	-0.005	0.005	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	ARRAY SEL X	26	2.4	-0.61	0.03	mA	1-3
F22	DATRETMODE	33	2.4	-0.005	0.005	mA	1-3
F23	MEMDATBUS17	44	2.4	-0.015	0.015	mA	1-3
F24	MEMDATBUS19	45					
F25	MEMDATBUS21	46	2.4	-0.015	0.015	mA	1-3
F26	MEMADRBUSA8	52	2.4	-0.2	0.02	mA	1-3
F27	MEMADRBUSA9	53					
F28	MEMADRBUSA10	54					
F29	MEMADRBUSA11	55	2.4	-0.2	0.02	mA	1-3
F30	MEMDATBUS0	56	2.4	-0.015	0.015	mA	1-3
F31	MEMDATBUS2	57					
F32	MEMDATBUS4	58					
F33	MEMDATBUS6	59	2.4	-0.015	0.015	mA	1-3
F34	MEMADRBUSA12	61	2.4	-0.2	0.02	mA	1-3
F35	MEMADRBUSA13	62					
F36	MEMADRBUSA14	63					
F37	MEMADRBUSA15	64	2.4	-0.2	0.02	mA	1-3
F38	MEMDATBUS8	67	2.4	-0.015	0.015	mA	1-3
F39	MEMDATBUS10	68					
F40	MEMDATBUS12	69	2.4	-0.015	0.015	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	MEMDATBUS14	70	2.4	-0.015	0.015	mA	1-3
F42	READ ENABLE	71	2.4	-0.03	0.03	mA	1-3
F43	INIT	86	2.4	-3.32	-1.42	mA	1-3
F44	MEMDATBUS16	94	2.4	-0.015	0.015	mA	1-3
F45	MEMDATBUS18	95					
F46	MEMDATBUS20	96	2.4	-0.015	0.015	mA	1-3
F47	MEMADRBUSB16	37	2.4	-0.2	0.02	mA	1-3
F48	MEMADRBUSB17	38					
F49	MEMADRBUSB0	39					
F50	MEMADRBUSB1	41					
F51	MEMADRBUSB2	42					
F52	MEMADRBUSB3	43					
F53	MEMADRBUSB4	47					
F54	MEMADRBUSB5	48					
F55	MEMADRBUSB6	49					
F56	MEMADRBUSB7	50					
F57	MEMADRBUSB8	89					
F58	MEMADRBUSB9	91					
F59	MEMADRBUSB10	92					
F60	MEMADRBUSB11	93	2.4	-0.2	0.02	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F61	MEMADRBUSB12	97	2.4	-0.2	0.02	mA	1-3
F62	MEMADRBUSB13	98					
F63	MEMADRBUSB14	99					
F64	MEMADRBUSB15	100	2.4	-0.2	0.02	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:19Z
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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F64
Power supply current limit	0.45 A	NA	F1-F64
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F64
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F64
low, (VL)	0.0 V	± 100 mV	F1-F64
Input current measurement	NA	± 11.0 uA	F1-F4
	NA	± 1.5 uA	F5-F8
	NA	± 11.0 uA	F9-F14
	NA	± 1.5 uA	F15-F18
	NA	± 0.5 uA	F19,F20
	NA	± 32.0 uA	F21
	NA	± 0.5 uA	F22
	NA	± 1.5 uA	F23-F25
	NA	± 11.0 uA	F26-F29
	NA	± 1.5 uA	F30-F33
	NA	± 11.0 uA	F34-F37
	NA	± 1.5 uA	F38-F41
	NA	± 3.0 uA	F42
	NA	± 95.0 uA	F43
	NA	± 1.5 uA	F44-F46
	NA	± 11.0 uA	F47-F64

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	MEMADRBUSA0	2	0.4	-0.2	0.02	mA	1-3
G2	MEMADRBUSA1	3					
G3	MEMADRBUSA2	4					
G4	MEMADRBUSA3	5	0.4	-0.2	0.02	mA	1-3
G5	MEMDATBUS1	6	0.4	-0.015	0.015	mA	1-3
G6	MEMDATBUS3	7					
G7	MEMDATBUS5	8					
G8	MEMDATBUS7	9	0.4	-0.015	0.015	mA	1-3
G9	MEMADRBUSA4	11	0.4	-0.2	0.02	mA	1-3
G10	MEMADRBUSA5	12					
G11	MEMADRBUSA6	13					
G12	MEMADRBUSA7	14					
G13	MEMADRBUSA16	15					
G14	MEMADRBUSA17	16	0.4	-0.2	0.02	mA	1-3
G15	MEMDATBUS9	17	0.4	-0.015	0.015	mA	1-3
G16	MEMDATBUS11	18					
G17	MEMDATBUS13	19					
G18	MEMDATBUS15	20	0.4	-0.015	0.015	mA	1-3
G19	WRITEDATPULS	22	0.4	-0.005	0.005	mA	1-3
G20	WRITECHCKBIT	23	0.4	-0.005	0.005	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
G21	ARRAY SEL X	26	0.4	-0.61	0.03	mA	1-3
G22	DATRETMODE	33	0.4	-0.005	0.005	mA	1-3
G23	MEMDATBUS17	44	0.4	-0.015	0.015	mA	1-3
G24	MEMDATBUS19	45					
G25	MEMDATBUS21	46	0.4	-0.015	0.015	mA	1-3
G26	MEMADRBUSA8	52	0.4	-0.2	0.02	mA	1-3
G27	MEMADRBUSA9	53					
G28	MEMADRBUSA10	54					
G29	MEMADRBUSA11	55	0.4	-0.2	0.02	mA	1-3
G30	MEMDATBUS0	56	0.4	-0.015	0.015	mA	1-3
G31	MEMDATBUS2	57					
G32	MEMDATBUS4	58					
G33	MEMDATBUS6	59	0.4	-0.015	0.015	mA	1-3
G34	MEMADRBUSA12	61	0.4	-0.2	0.02	mA	1-3
G35	MEMADRBUSA13	62					
G36	MEMADRBUSA14	63					
G37	MEMADRBUSA15	64	0.4	-0.2	0.02	mA	1-3
G38	MEMDATBUS8	67	0.4	-0.015	0.015	mA	1-3
G39	MEMDATBUS10	68					
G40	MEMDATBUS12	69	0.4	-0.015	0.015	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	MEMDATBUS14	70	0.4	-0.015	0.015	mA	1-3
G42	READ ENABLE	71	0.4	-0.03	0.03	mA	1-3
G43	INIT	86	0.4	-3.32	-1.42	mA	1-3
G44	MEMDATBUS16	94	0.4	-0.015	0.015	mA	1-3
G45	MEMDATBUS18	95					
G46	MEMDATBUS20	96	0.4	-0.015	0.015	mA	1-3
G47	MEMADRBUSB16	37	0.4	-0.2	0.02	mA	1-3
G48	MEMADRBUSB17	38					
G49	MEMADRBUSB0	39					
G50	MEMADRBUSB1	41					
G51	MEMADRBUSB2	42					
G52	MEMADRBUSB3	43					
G53	MEMADRBUSB4	47					
G54	MEMADRBUSB5	48					
G55	MEMADRBUSB6	49					
G56	MEMADRBUSB7	50					
G57	MEMADRBUSB8	89					
G58	MEMADRBUSB9	91					
G59	MEMADRBUSB10	92					
G60	MEMADRBUSB11	93	0.4	-0.2	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G61	MEMADRBUSB12	97	0.4	-0.2	0.02	mA	1-3
G62	MEMADRBUSB13	98					
G63	MEMADRBUSB14	99					
G64	MEMADRBUSB15	100	0.4	-0.2	0.02	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G64
Power supply current limit	0.45 A	NA	G1-G64
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G64
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G64
low, (VL)	0.0 V	± 100 mV	G1-G64
Input current measurement	NA	± 11.0 uA	G1-G4
	NA	± 1.5 uA	G5-G8
	NA	± 11.0 uA	G9-G14
	NA	± 1.5 uA	G15-G18
	NA	± 0.5 uA	G19,G20
	NA	± 32.0 uA	G21
	NA	± 0.5 uA	G22
	NA	± 1.5 uA	G23-G25
	NA	± 11.0 uA	G26-G29
	NA	± 1.5 uA	G30-G33
	NA	± 11.0 uA	G34-G37
	NA	± 1.5 uA	G38-G41
	NA	± 3.0 uA	G42
	NA	± 95.0 uA	G43
	NA	± 1.5 uA	G44-G46
	NA	± 11.0 uA	G47-G64

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	MEMDATBUS1	6	0.4	2.4	4.5	V	1-4
H2	MEMDATBUS3	7					
H3	MEMDATBUS5	8					
H4	MEMDATBUS7	9					
H5	MEMDATBUS9	17					
H6	MEMDATBUS11	18					
H7	MEMDATBUS13	19					
H8	MEMDATBUS15	20					
H9	MEMDATBUS17	44					
H10	MEMDATBUS19	45					
H11	MEMDATBUS21	46					
H12	MEMDATBUS0	56					
H13	MEMDATBUS2	57					
H14	MEMDATBUS4	58					
H15	MEMDATBUS6	59					
H16	MEMDATBUS8	67					
H17	MEMDATBUS10	68					
H18	MEMDATBUS12	69					
H19	MEMDATBUS14	70					
H20	MEMDATBUS16	94	0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	MEMDATBUS18	95	0.4	2.4	4.5	V	1-4
H22	MEMDATBUS20	96	0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H22
Power supply current limit	0.45 A	NA	H1-H22
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1-H22
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H22 H1-H22
Output voltage measurement	NA	± 105 mV	H1-H22

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

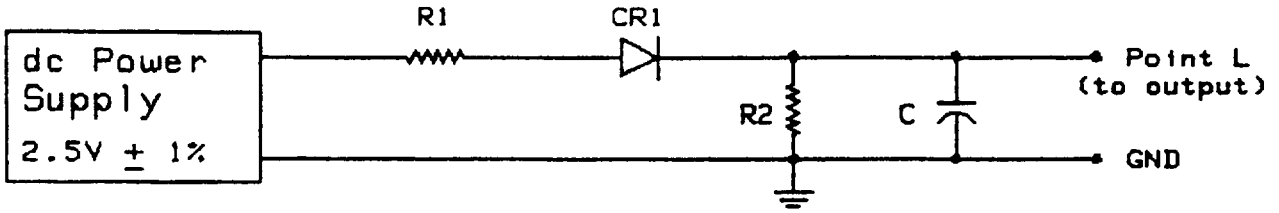
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
6,7,8,9,17,18,19, 20,44,45,46,56,57, 58,59,67,68,69,70, 94,95,96	6.2 K	6.0	0.5
77,79,80,84	NA	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	MEMDATBUS1	6	6.0	0.0	0.5	V	1-4
J2	MEMDATBUS3	7					
J3	MEMDATBUS5	8					
J4	MEMDATBUS7	9					
J5	MEMDATBUS9	17					
J6	MEMDATBUS11	18					
J7	MEMDATBUS13	19					
J8	MEMDATBUS15	20					
J9	MEMDATBUS17	44					
J10	MEMDATBUS19	45					
J11	MEMDATBUS21	46					
J12	TYPE 0	77					
J13	SIZE 0	79					
J14	SIZE 1	80					
J15	HERE	84					
J16	MEMDATBUS0	56					
J17	MEMDATBUS2	57					
J18	MEMDATBUS4	58					
J19	MEMDATBUS6	59					
J20	MEMDATBUS8	67					
J21	MEMDATBUS10	68	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J22	MEMDATBUS12	69	6.0	0.0	0.5	V	1-4
J23	MEMDATBUS14	70					
J24	MEMDATBUS16	94					
J25	MEMDATBUS18	95					
J26	MEMDATBUS20	96	6.0	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:19Z
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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J26
Power supply current limit	0.45 A	NA	J1-J26
Forcing current, (Iol)	6.0 mA	± 600 uA	J1-J26
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J26 J1-J26
Output voltage measurement	NA	± 25 mV	J1-J26

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
High level							
K1	TYPE 0	77	2.4	-0.01	0.01	mA	1-3
K2	SIZE 0	79					
K3	SIZE 1	80					
K4	HERE	84	2.4	-0.01	0.01	mA	1-3
Low level							
K5	TYPE 0	77	2.4	-0.01	0.01	mA	1-3
K6	SIZE 0	79					
K7	SIZE 1	80					
K8	HERE	84	2.4	-0.01	0.01	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K8
Power supply current limit	0.45 A	NA	K1-K8
Forcing voltage, (Voh) (Vol)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 K1-K4 K5-K8
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K8 K1-K8
Output current measurement	 NA NA	 ± 1 uA ± 1 uA	 K1-K4 K5-K8

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	MEMADRBUSA0	2	-18.0	-1.5	0.0	V	1-3
L2	MEMADRBUSA1	3					
L3	MEMADRBUSA2	4					
L4	MEMADRBUSA3	5					
L5	MEMADRBUSA4	11					
L6	MEMADRBUSA5	12					
L7	MEMADRBUSA6	13					
L8	MEMADRBUSA7	14					
L9	MEMADRBUSA16	15					
L10	MEMADRBUSA17	16	-18.0	-1.5	0.0	V	1-3
L11	WRITEDATPULS	22	-18.0	-1.2	0.0	V	1-3
L12	WRITECHCKBIT	23					
L13	ARRAY SEL X	26					
L14	DATRETMODE	33	-18.0	-1.2	0.0	V	1-3
L15	MEMADRBUSB16	37	-18.0	-1.5	0.0	V	1-3
L16	MEMADRBUSB17	38					
L17	MEMADRBUSB0	39					
L18	MEMADRBUSB1	41					
L19	MEMADRBUSB2	42					
L20	MEMADRBUSB3	43	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	MEMADRBUSB4	47	-18.0	-1.5	0.0	V	1-3
L22	MEMADRBUSB5	48					
L23	MEMADRBUSB6	49					
L24	MEMADRBUSB7	50					
L25	MEMADRBUSA8	52					
L26	MEMADRBUSA9	53					
L27	MEMADRBUSA10	54					
L28	MEMADRBUSA11	55					
L29	MEMADRBUSA12	61					
L30	MEMADRBUSA13	62					
L31	MEMADRBUSA14	63					
L32	MEMADRBUSA15	64	-18.0	-1.5	0.0	V	1-3
L33	READ ENABLE	71	-18.0	-1.2	0.0	V	1-3
L34	MEMADRBUSB8	89	-18.0	-1.5	0.0	V	1-3
L35	MEMADRBUSB9	91					
L36	MEMADRBUSB10	92					
L37	MEMADRBUSB11	93					
L38	MEMADRBUSB12	97					
L39	MEMADRBUSB13	98					
L40	MEMADRBUSB14	99	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L41	MEMADRBUSB15	100	-18.0	-1.5	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L41
Power supply current limit	NA	NA	L1-L41
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L41
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L41 L1-L41
Input voltage measurement	NA NA NA NA NA	± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV	L1-L10 L11-L14 L15-L32 L33 L34-L41

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	2.8	5.2	uF	1-4
M2	PSFC	51	2.8	5.2	uF	1-4
M3	PSFC	36	7.4	13.8	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1-M3
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1-M3
Capacitance measurement	NA NA	± 120 nF ± 0.32 uF	M1-M2 M3

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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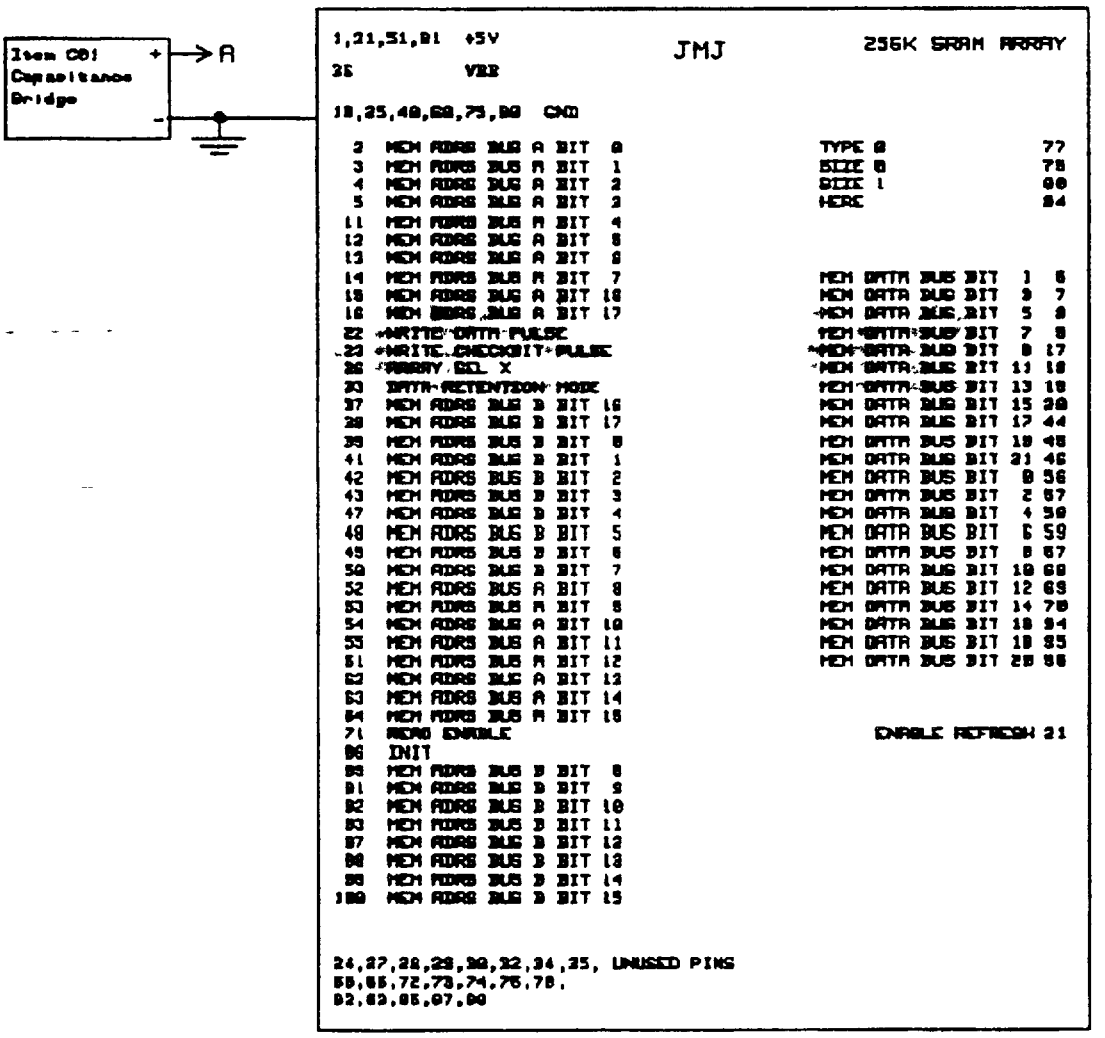


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N1	Icc	1	100	300	mA	1-3
N2	VBB	36	200	1500	mA	1-3
N3	VBB (backup)	36	0	20	mA	1-3

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NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N1-N3
Power supply current limit	0.45 A	NA	N1-N3
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N1-N3 N1-N3
Power supply current measurement	 NA NA NA	 ± 10 mA ± 65 mA ± 1 mA	 N1 N2 N3

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	0.45 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

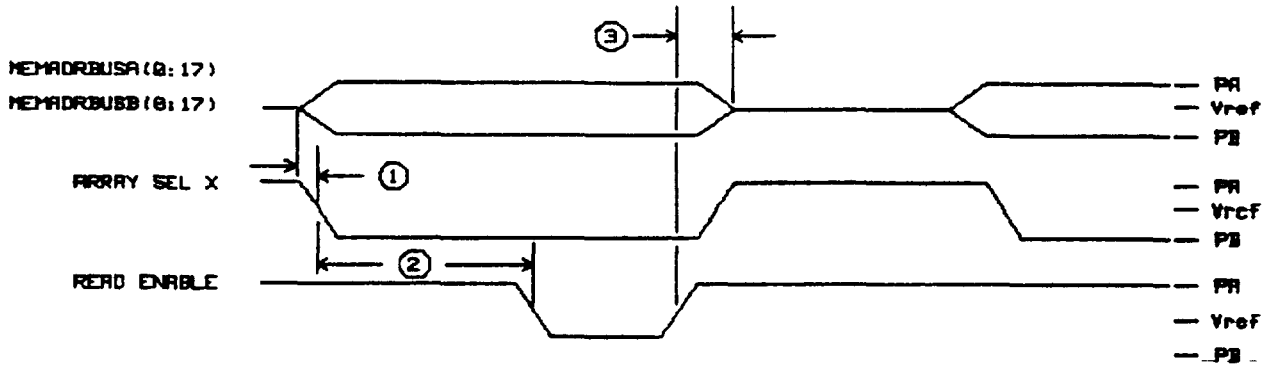
(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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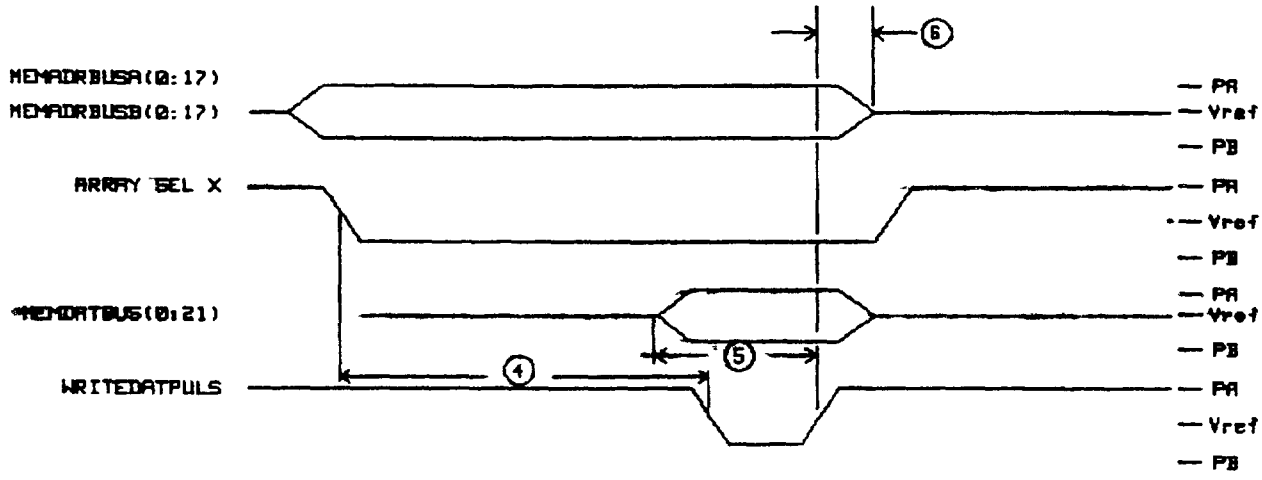


NOTES:

1. (1) MEMADRBUSA(0:17) and MEMADRBUSB(0:17) to ARRAY SEL X. Setup time: 10 nS minimum.
2. (2) ARRAY SEL X to READ ENABLE. Setup time: 85 nS minimum.
3. (3) READ ENABLE to MEMADRBUSA(0:17) and MEMADRBUSB(0:17). Setup time: 0 nS minimum.

FIGURE 11a. Input timing parameters for table XV.

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NOTES:

1. (4) ARRAY SEL X to WRITEDATPULS. Setup time: 83 nS minimum.

2. (5) MEMDATBUS(0:21) to WRITEDATPULS. Setup time: 74 nS minimum.

3. (6) WRITEDATPULS to MEMADRBUSA(0:17) and MEMADRBUSB(0:17). Setup time: 8 nS minimum.

FIGURE 11b. Input timing parameters for table XV.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	0.45 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

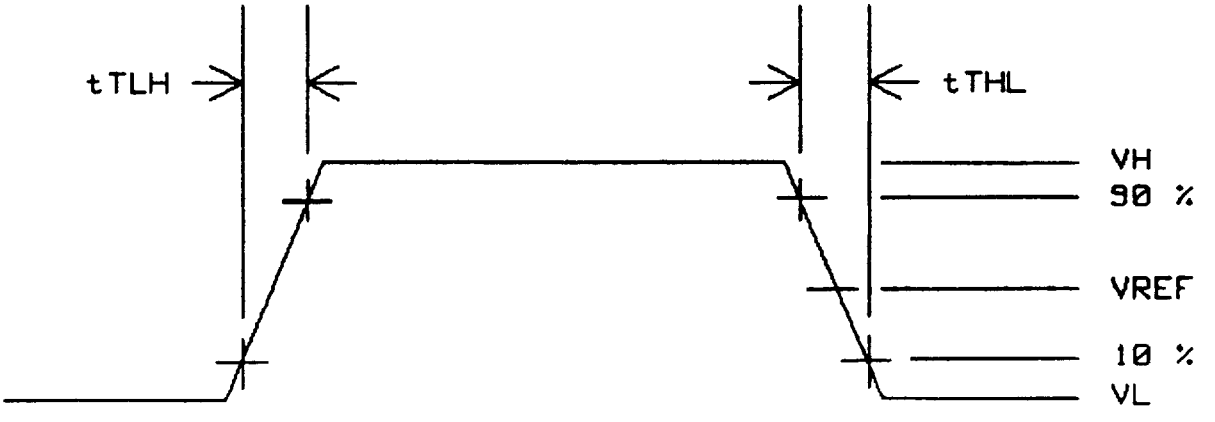
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	MEMADRBUSA0 to ARRAY SEL X	10	10	10	-	-	-	nS	12a, 12b
1	tsu	MEMADRBUSA1 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA2 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA3 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA4 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA5 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA6 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA7 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA8 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA9 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA10 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA11 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA12 to ARRAY SEL X	10	10	10	-	-	-	nS	12a, 12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	MEMADRBUSA13 to ARRAY SEL X	10	10	10	-	-	-	nS	12a, 12b
1	tsu	MEMADRBUSA14 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA15 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA16 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSA17 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB0 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB1 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB2 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB3 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB4 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB5 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB6 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB7 to ARRAY SEL X	10	10	10	-	-	-	nS	12a, 12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	MEMADRBUSB8 to ARRAY SEL X	10	10	10	-	-	-	nS	12a, 12b
1	tsu	MEMADRBUSB9 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB10 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB11 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB12 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB13 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB14 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB15 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB16 to ARRAY SEL X	10	10	10	-	-	-	nS	
1	tsu	MEMADRBUSB17 to ARRAY SEL X	10	10	10	-	-	-	nS	
2	tsu	ARRAY SEL X to READ ENABLE	85	85	85	-	-	-	nS	
3	tpd	ARRAY SEL X to MEMDATBUS0	-	-	-	180	180	180	nS	12a, 12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	ARRAY SEL X to MEMDATBUS1	-	-	-	180	180	180	nS	12a, 12b
3	tpd	ARRAY SEL X to MEMDATBUS2	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS3	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS4	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS5	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS6	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS7	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS8	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS9	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS10	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS11	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS12	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS13	-	-	-	180	180	180	nS	12a, 12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	ARRAY SEL X to MEMDATBUS14	-	-	-	180	180	180	nS	12a, 12b
3	tpd	ARRAY SEL X to MEMDATBUS15	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS16	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS17	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS18	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS19	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS20	-	-	-	180	180	180	nS	
3	tpd	ARRAY SEL X to MEMDATBUS21	-	-	-	180	180	180	nS	
4	tsu	READ ENABLE to MEMADRBUSA0	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA1	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA2	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA3	0	0	0	-	-	-	nS	12a, 12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tsu	READ ENABLE to MEMADRBUSA4	0	0	0	-	-	-	nS	12a, 12b
4	tsu	READ ENABLE to MEMADRBUSA5	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA6	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA7	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA8	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA9	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA10	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA11	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA12	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA13	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA14	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA15	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSA16	0	0	0	-	-	-	nS	12a, 12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tsu	READ ENABLE to MEMADRBUSA17	0	0	0	-	-	-	nS	12a, 12b
4	tsu	READ ENABLE to MEMADRBUSB0	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB1	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB2	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB3	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB4	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB5	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB6	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB7	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB8	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB9	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB10	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB11	0	0	0	-	-	-	nS	12a, 12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tsu	READ ENABLE to MEMADRBUSB12	0	0	0	-	-	-	nS	12a, 12b
4	tsu	READ ENABLE to MEMADRBUSB13	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB14	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB15	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB16	0	0	0	-	-	-	nS	
4	tsu	READ ENABLE to MEMADRBUSB17	0	0	0	-	-	-	nS	
5	tpw	ARRAY SEL X	156	156	156	-	-	-	nS	
6	tsu	ARRAY SEL X to WRITEDATPULS	83	83	83	-	-	-	nS	
7	tsu	MEMDATBUS0 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS1 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS2 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS3 to WRITEDATPULS	74	74	74	-	-	-	nS	12a, 12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tsu	MEMDATBUS4 to WRITEDATPULS	74	74	74	-	-	-	nS	12a, 12b
7	tsu	MEMDATBUS5 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS6 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS7 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS8 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS9 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS10 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tus	MEMDATBUS11 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS12 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS13 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS14 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS15 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS16 to WRITEDATPULS	74	74	74	-	-	-	nS	12a, 12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tsu	MEMDATBUS17 to WRITEDATPULS	74	74	74	-	-	-	nS	12a, 12b
7	tsu	MEMDATBUS18 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS19 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS20 to WRITEDATPULS	74	74	74	-	-	-	nS	
7	tsu	MEMDATBUS21 to WRITEDATPULS	74	74	74	-	-	-	nS	
8	tpw	WRITEDATPULS	50	50	50	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA0	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA1	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA2	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA3	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA4	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA5	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA6	8	8	8	-	-	-	ns	12a, 12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tsu	WRITEDATPULS to MEMADRBUSA7	8	8	8	-	-	-	nS	12a, 12b
9	tsu	WRITEDATPULS to MEMADRBUSA8	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA9	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA10	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA11	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA12	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA13	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA14	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA15	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA16	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSA17	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB0	8	8	8	-	-	-	nS	12a, 12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tsu	WRITEDATPULS to MEMADRBUSB1	8	8	8	-	-	-	nS	12a, 12b
9	tsu	WRITEDATPULS to MEMADRBUSB2	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS tO MEMADRBUSB3	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB4	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB5	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB6	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB7	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB8	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB9	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB10	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB11	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB12	8	8	8	-	-	-	nS	12a, 12b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
9	tsu	WRITEDATPULS to MEMADRBUSB13	8	8	8	-	-	-	nS	12a, 12b
9	tsu	WRITEDATPULS to MEMADRBUSB14	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB15	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB16	8	8	8	-	-	-	nS	
9	tsu	WRITEDATPULS to MEMADRBUSB17	8	8	8	-	-	-	nS	12a, 12b

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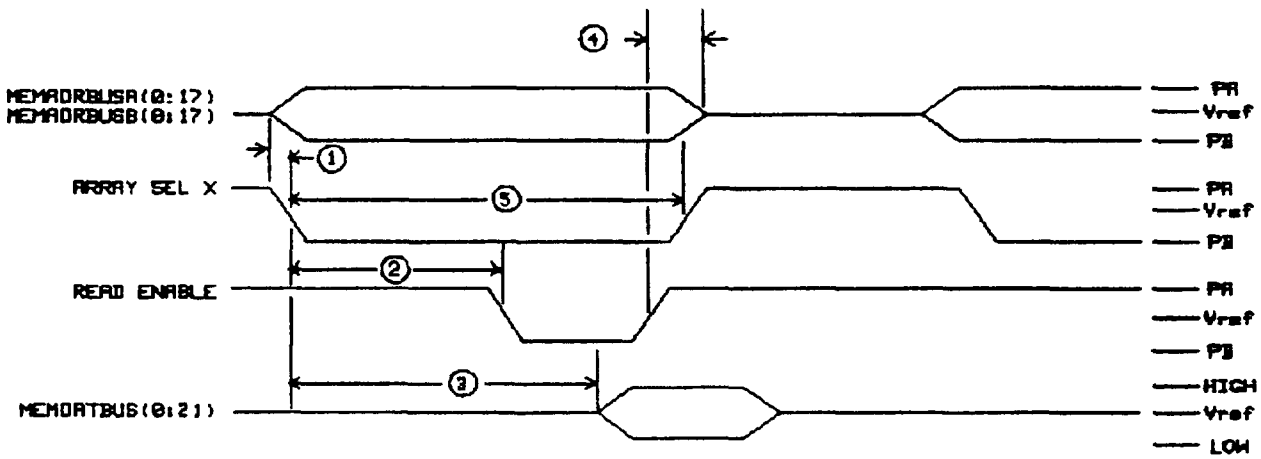


FIGURE 12a. Dynamic waveforms for table XVII.

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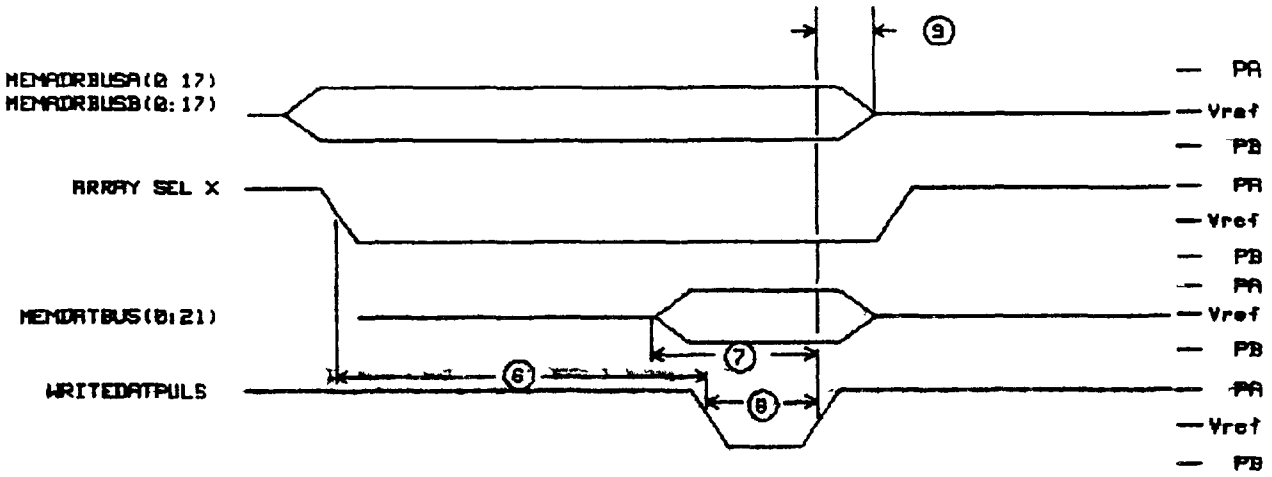


FIGURE 12b. Dynamic waveforms for table XVII.

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TABLE XVIII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R1	MEMADRBUSA0	2	2	50	pF	1-4
R2	MEMADRBUSA1	3				
R3	MEMADRBUSA2	4				
R4	MEMADRBUSA3	5				
R5	MEMADRBUSA4	11				
R6	MEMADRBUSA5	12				
R7	MEMADRBUSA6	13				
R8	MEMADRBUSA7	14				
R9	MEMADRBUSA16	15				
R10	MEMADRBUSA17	16				
R11	WRITEDATPULS	22				
R12	WRITECHCKBIT	23	2	50	pF	1-4
R13	ARRAY SEL X	26	4	75	pF	1-4
R14	DATRETMODE	33	2	50	pF	1-4
R15	MEMADRBUSB16	37				
R16	MEMADRBUSB17	38				
R17	MEMADRBUSB0	39				
R18	MEMADRBUSB1	41				
R19	MEMADRBUSB2	42				
R20	MEMADRBUSB3	43	2	50	pF	1-4

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TABLE XVIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R21	MEMADRBUSB4	47	2	50	pF	1-4
R22	MEMADRBUSB5	48				
R23	MEMADRBUSB6	49				
R24	MEMADRBUSB7	50				
R25	MEMADRBUSA8	52				
R26	MEMADRBUSA9	53				
R27	MEMADRBUSA10	54				
R28	MEMADRBUSA11	55				
R29	MEMADRBUSA12	61				
R30	MEMADRBUSA13	62				
R31	MEMADRBUSA14	63				
R32	MEMADRBUSA15	64	2	50	pF	1-4
R33	READ ENABLE	71	6	150	pF	1-4
R34	INIT	86	4	75	pF	1-4
R35	MEMADRBUSB8	89	2	50	pF	1-4
R36	MEMADRBUSB9	91				
R37	MEMADRBUSB10	92				
R38	MEMADRBUSB11	93				
R39	MEMADRBUSB12	97				
R40	MEMADRBUSB13	98	2	50	pF	1-4

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TABLE XVIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	MEMADRBUSB14	99	2	50	pF	1-4
R42	MEMADRBUSB15	100	2	50	pF	1-4
R43	MEMDATBUS1	6	4	50	pF	1-4
R44	MEMDATBUS3	7				
R45	MEMDATBUS5	8				
R46	MEMDATBUS7	9				
R47	MEMDATBUS9	17				
R48	MEMDATBUS11	18				
R49	MEMDATBUS13	19				
R50	MEMDATBUS15	20				
R51	MEMDATBUS17	44				
R52	MEMDATBUS19	45				
R53	MEMDATBUS21	46				
R54	MEMDATBUS0	56				
R55	MEMDATBUS2	57				
R56	MEMDATBUS4	58				
R57	MEMDATBUS6	59				
R58	MEMDATBUS8	67				
R59	MEMDATBUS10	68				
R60	MEMDATBUS12	69	4	50	pF	1-4

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TABLE XVIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R61	MEMDATBUS14	70	4	50	pF	1-4
R62	MEMDATBUS16	94				
R63	MEMDATBUS18	95				
R64	MEMDATBUS20	96	4	50	pF	1-4

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NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R64
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R64
Capacitance measurement	NA	± 2.4 pF	R1-R12
	NA	± 3.55 pF	R13
	NA	± 2.4 pF	R14-R32
	NA	± 7.2 pF	R33
	NA	± 3.55 pF	R34
	NA	± 2.4 pF	R35-R42
	NA	± 2.3 pF	R43-R64

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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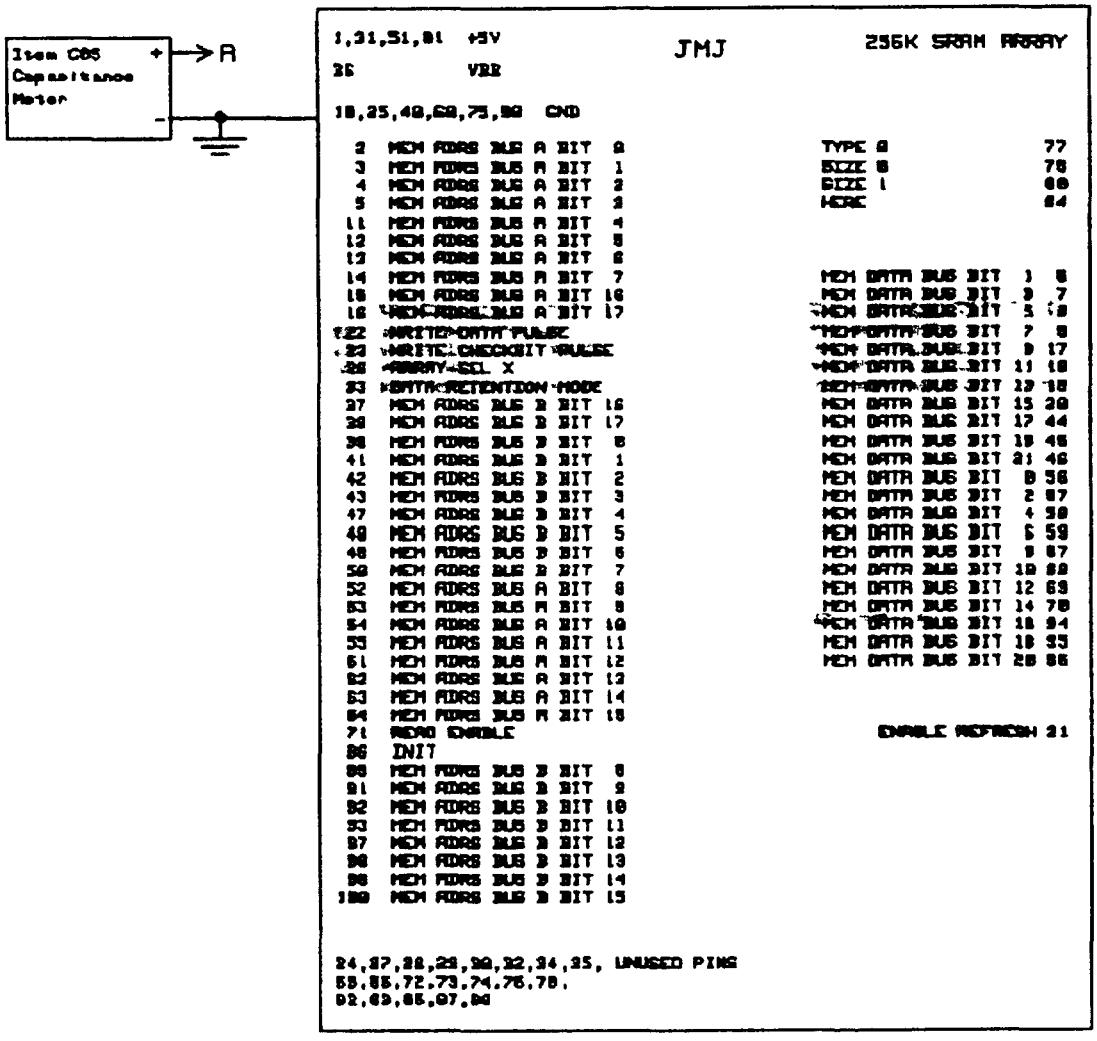


FIGURE 13. Input capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4
T3	PSTA	36	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1-T3
Power supply current limit	0.45 A	NA	T1-T3
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1-T3 T1-T3
Power supply transient amplitude measurement	NA	± 20 mV	T1-T3

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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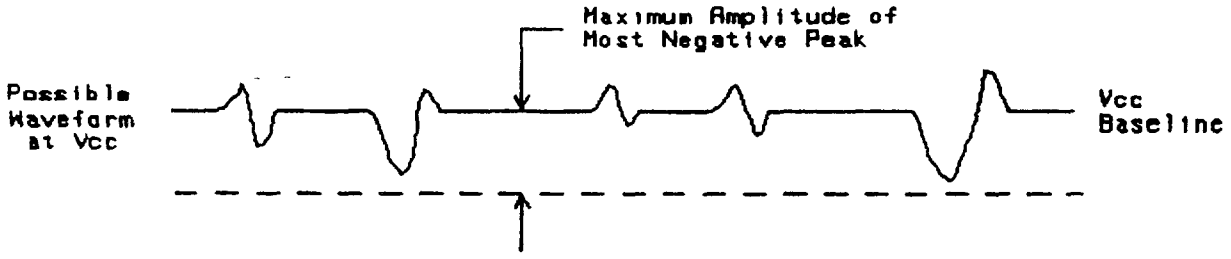
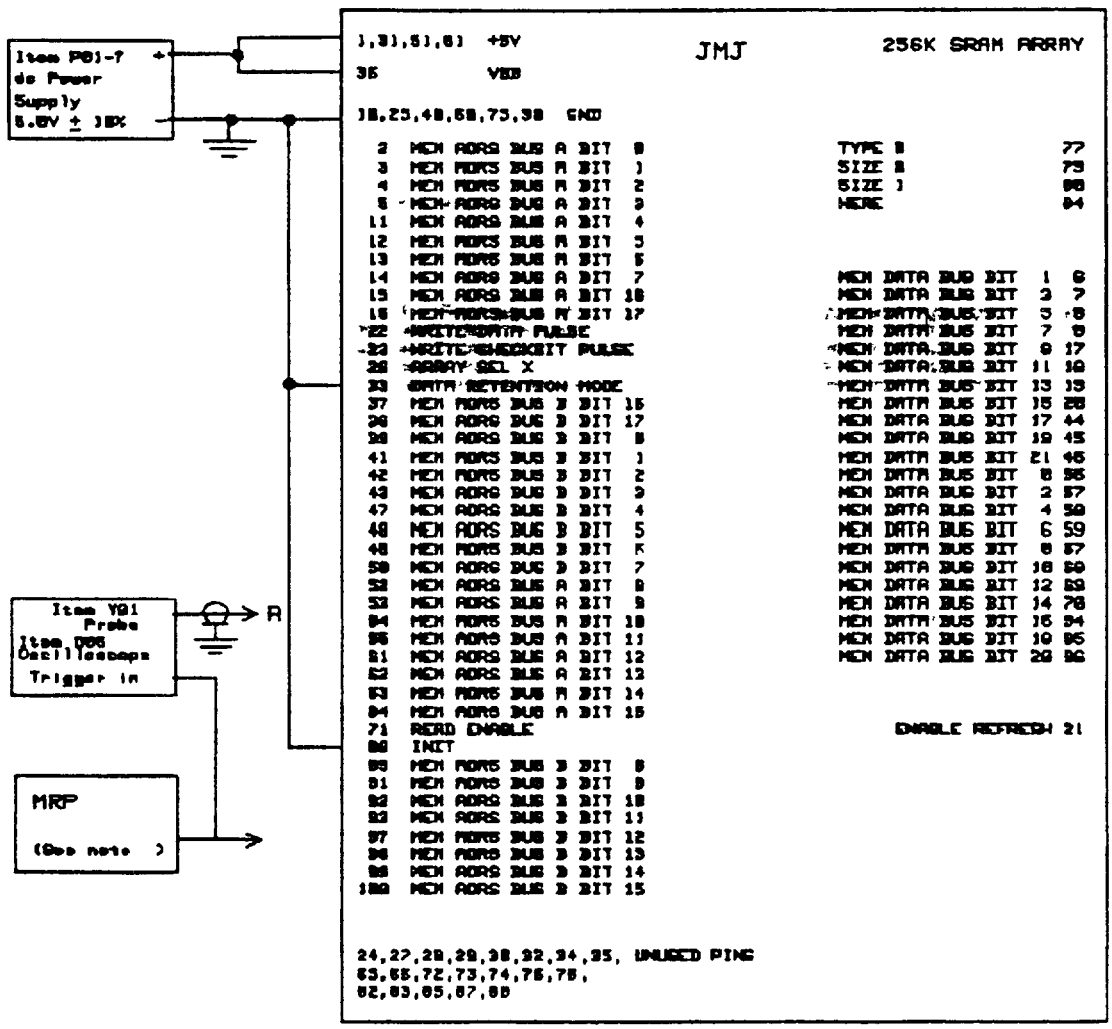


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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